

Oakwood AI Innovation Lab — Meeting Briefing & Implementation Companion

A polished decision brief for a 10-seat AI production studio where Math & Computer Science students learn modern AI by building, evaluating, shipping, and presenting real work.

Executive thesis. Oakwood does not need another ordinary computer lab. It needs a visible AI production studio: a room with capable student seats, shared local model infrastructure, a demo wall, governance habits, and a 90-day path to the first public proof.

10

student builder seats

4

budget paths

90

day launch runway

AI

local plus cloud
model

The lab is a production studio, not a row of upgraded machines.

The center of the proposal is a student outcome: every participant leaves with visible artifacts that can be shown to faculty, recruiters, grant reviewers, and community partners.

CORE POSITION

Build the room around shipping.

The lab should be designed for weekly progress: app prototypes, local model demos, responsible-AI artifacts, short demo videos, and public showcases. Hardware matters because it makes that cadence real.

DECISION FRAME

Choose the smallest tier that can produce proof.

The \$50K Applied Builder path is the most balanced starting point. The \$20K pod proves the thesis faster, while the \$100K and \$200K paths support research, recruitment, and grant visibility from launch.

Use

AI fluency, research, analysis, coding, and presentation.

Build

Apps, agents, automations, dashboards, voice, and vision tools.

Govern

Evaluation, data handling, approval gates, audits, and policy drafts.

Meeting objective. Confirm the available funding range, the room constraints, the program format, the Year 1 faculty lead, and the first measurable definition of success. Those five answers determine the next procurement memo and pilot curriculum.

AI has moved from a topic students study to a production skill employers expect.

The advantage in 2026 is not generic exposure to tools. It is the ability to build a useful system, explain its limits, evaluate its behavior, and ship it with care.

PRESSURE	LAB RESPONSE	STUDENT EVIDENCE
AI velocity	Weekly build cadence with modern coding, model, and deployment tools.	Live apps, GitHub history, demo recordings, and technical briefs.
Workforce demand	Local inference, cloud workflows, data discipline, and evaluation practice.	Portfolio projects aligned to internships, research, and Huntsville industry needs.
Equity of access	Shared high-memory nodes and reliable seats, so capability does not depend on a student's personal machine.	A transparent cohort process and consistent access to serious compute.
Trust and safety	Governance taught as a craft, not a warning slide.	Data labels, approval gates, test reports, red-team notes, and audit trails.

HBCU LEADERSHIP

Oakwood can frame the lab as leadership, not catch-up: a visible place where students become builders of the systems shaping their communities.

HUNTSVILLE FIT

Defense, aerospace, health, biotech, and civic employers increasingly need graduates who can pair technical skill with judgment.

TIMING

A first cohort, demo day, and follow-on grant narrative can all be produced inside one semester if the room and program format are chosen now.

Dr. Smith's ask is practical: what will actually be useful?

The brief should respect the department's need for usable guidance, not a catalog of machines. Start with outcomes, then map the room, hardware, governance, and curriculum to those outcomes.

KNOWN DIRECTION

- The department is considering computer-lab upgrades with an AI focus.
- Funding has been identified and needs a defensible recommendation.
- The useful answer is a complete model: room, stack, governance, training, and student outcomes.
- The conversation should confirm scope before locking a tier.

FIT FOR DR. SMITH & MATH/CS

- Mission-aligned STEM access and HBCU excellence.
- Rigor without jargon: every technical claim should translate to student value.
- Faculty time reality: a bridge week and mentor model reduce launch friction.
- Responsible AI as a department strength, not an afterthought.

Conversation posture. Listen first. Confirm what success looks like for the department, then use the briefing to narrow choices. The best outcome is agreement on a tier or agreement on the next diagnostic deliverable.

Room

Dimensions, power, HVAC, network, AV, furniture, and security constraints.

Format

Course, certificate, club, summer bridge, faculty development, or combined model.

Proof

Portfolios, internships, demo day, grant leverage, retention, and faculty confidence.

Four capabilities organize the lab and keep the equipment tied to learning.

Every student should move through the same progression: use AI well, build with it, run models directly, and govern what they ship.

Use

AI fluency

Prompting, research, analysis, source checking, design, coding assistance, and presentations.

Build

Creation

Apps, agents, automations, RAG systems, dashboards, voice, vision, and campus tools.

Run

Infrastructure

Ollama, LM Studio, vLLM, embeddings, quantization, latency, privacy, and cost tradeoffs.

Govern

Responsibility

Evaluation, red-teaming, prompt-injection defense, bias checks, policy drafts, and audits.

EIGHT-MODULE SPINE

1. AI fluency
2. Build week
3. Local models
4. Agents and automation
5. AI security and governance
6. Data, dashboards, and story
7. Voice, vision, and media
8. Capstone studio

ROOM ZONES

Student Build Zone: ten seats for pair work, reviews, and daily production.

Local Model Bench: shared high-memory nodes for frontier open models.

Demo Wall: weekly showcases, visitor presentations, and critique.

Studio Corner: camera, lighting, and mics for portfolio-ready demos.

Edge AI Bench: Jetson devices, sensors, robotics kits, and physical AI.

The hardware model is modular: reliable seats, shared local AI, storage, studio output, and edge devices.

The lab should be quoted as a system. Final procurement should use current education pricing and a facilities check before purchase.

COMPONENT	ROLE IN THE LAB	PLANNING GUIDANCE
Student seats	Daily workstations for coding, AI-assisted build work, design, data, and demo production.	Prioritize reliability, 64-128 GB memory where possible, fast storage, and standard x86 compatibility for broad tooling.
DGX Spark-class node	Shared high-memory local inference, model serving, evaluation, and local-model teaching.	Use as a shared bench. Arm-based tooling is strong, but keep student seats compatible with common desktop workflows.
Framework Desktop AI Max+ class	High-memory daily AI workstation with strong local-model economics.	Most defensible for the recommended 10-seat Applied Builder tier.
Mac Studio class	Media, voice, vision, creative pipelines, and premium studio work.	Best in hybrid or higher tiers when Apple workflows matter.
Jetson AGX / edge kits	Robotics, vision sensors, low-latency inference, and physical AI demonstrations.	Adds a visible hardware hook and broadens student interest beyond software.
Network, NAS, backup	Shared datasets, model weights, snapshots, access control, and demo-day reliability.	10 GbE minimum; 25 GbE+ for higher tiers. Include backup and device management from day one.

Procurement note. Planning prices in the source material are mid-May 2026 estimates. Memory, GPU, and education pricing can move quickly; every tier should be re-quoted before approval.

Local

Teaches privacy-bounded work, model serving, latency, quantization, and predictable inference economics.

Cloud

Handles occasional fine-tuning, larger batch jobs, and credits-backed experiments beyond the room.

Hybrid

Gives the department a practical mix instead of forcing one vendor ecosystem.

Four paths scale ambition without changing the identity of the lab.

Each tier should still produce a student build space, a local model experience, a demo surface, and a governance workflow. The difference is depth and durability.

STARTER POD

\$20K

Pilot fast. Ten capable seats, one shared local AI node, 10 GbE, NAS, demo display, and basic capture gear.

Best for proving the thesis inside one semester.

APPLIED BUILDER • RECOMMENDED

\$50K

Balanced launch. Ten 128 GB Framework-class seats, one to two DGX Spark-class nodes, stronger storage, demo wall, studio kit, and edge starters.

Best blend of student access and credible local AI.

RESEARCH STUDIO

\$100K

Grant credible. Premium seats, multiple local nodes, shared GPU server, 25 GbE backbone, enterprise NAS, expanded AV, and Jetson AGX class devices.

Best for capstones and faculty research.

FLAGSHIP CENTER

\$200K

Signature hub. Elite seats, DGX cluster, dedicated GPU server, 100 GbE backbone, all-flash storage, full studio AV, edge fleet, and security logging.

Best for recruitment, grants, and partnerships.

DECISION	USE THIS PATH WHEN...	NEXT STEP
\$20K	The department needs a fast proof before a larger purchase.	Define a one-semester pilot and schedule the first demo day now.
\$50K	The goal is a serious 10-seat launch with local AI and production-grade demos.	Prepare line-item vendor quotes and an 8-week pilot curriculum.
\$100K	The lab must support capstones, faculty projects, and a stronger grant narrative.	Add facilities review, enterprise storage, and research workflow design.
\$200K	The university wants a signature center that can recruit, host partners, and anchor press-worthy demonstrations.	Run a full room, procurement, security, AV, and sustainability plan.

All paths should include a 10–15% contingency for shipping, cabling, peripherals, installation, and quote movement.

The lab should operate as a simple, teachable system.

Student workstations handle daily creation. Shared nodes run larger local models. Cloud credits cover overflow. Storage, backup, and policy make the work repeatable.

OPERATING STACK

Build: GitHub, VS Code/Cursor-class editors, Python, JavaScript/TypeScript, containers, Vercel, Cloudflare, Supabase or equivalent.

Model work: Ollama, LM Studio, vLLM, Hugging Face, embeddings, RAG, evaluation harnesses, and model cards.

Governance: dataset labels, project briefs, approval gates, test logs, prompt-injection checks, and demo review.

Publication: demo wall, short videos, one-page technical briefs, and portfolio pages.

LAUNCH RHYTHM

Faculty bridge: one week before cohort launch, focused on vocabulary, tools, classroom boundaries, and evaluation rubrics.

Student cohort: first cohort selected with a transparent rubric across class years and backgrounds.

Peer mentors: students help operate the lab by month three.

Weekly ritual: build review, responsible-AI check, demo recording, and next sprint plan.

Recommended default. Local-first for student and campus-sensitive work; cloud-burst for experiments that exceed the local memory or training budget. This keeps costs predictable while still giving students modern cloud fluency.

INFRASTRUCTURE ITEM	MINIMUM	HIGHER-TIER TARGET
Switching	10 GbE access for lab systems.	25 GbE or 100 GbE backbone for shared model and storage traffic.
Storage	NAS with snapshots and student project folders.	Enterprise NAS with cache, access control, backups, and dataset versioning.
Power/HVAC	Confirm outlets, circuits, and thermal headroom before purchase.	Facilities review for 4-5 kW peak loads in research or flagship tiers.
Security	Named accounts, backups, and device management.	SSO, audit logs, security appliance, and formal incident workflow.

Responsibility is part of the product students learn to build.

The lab should make responsible AI visible in the workflow: every project has a data posture, evaluation plan, human review point, and record of what changed.

GOVERNANCE PRACTICES

- Classify datasets before they touch a model.
- Keep human approval for sensitive actions such as email, database writes, payments, and publication.
- Run hallucination, bias, safety, and prompt-injection checks before demos.
- Document model choice, known limits, data sources, and evaluation results.
- Create a student-data handling policy before the first cohort begins.

SUCCESS METRICS

- Number of shipped student artifacts per cohort.
- Demo-day attendance and partner follow-up.
- Internship and portfolio outcomes that cite the lab.
- Faculty confidence before and after the bridge.
- Grant, sponsor, and partnership leads created from the pilot.

Policy

Lab use policy, data taxonomy, approval gates, and incident workflow.

Artifacts

Model cards, evaluation notes, demo scripts, technical briefs, and repositories.

Reporting

A semester dashboard that turns activity into follow-on funding evidence.

Why this matters at an HBCU. Oakwood graduates will build and evaluate systems that affect health, education, finance, civic services, and community life. Teaching governance as a craft is both a competitive advantage and a moral one.

Leave the conversation with enough clarity to write the next memo.

The meeting should narrow uncertainty. The strongest close is not a broad promise; it is a dated next deliverable tied to the tier and format Dr. Smith prefers.

FIVE QUESTIONS TO CONFIRM

1. What funding range is available, what source funds it, and is there a use-by date?
2. Which room is likely, and what are its power, network, HVAC, furniture, and security constraints?
3. What format should the lab support first: course, certificate, club, summer bridge, or faculty development?
4. Who will provide Year 1 faculty oversight, and who could become a second campus champion?
5. How should success be measured after the first semester?

DECISION POINTS

- Pick a target tier or request a room-and-procurement diagnostic.
- Choose whether the first cohort is credit-bearing or pilot-based.
- Decide how much local AI is required on day one.
- Approve a governance baseline before student projects begin.
- Set the next deliverable date before the meeting ends.

IF SHE ASKS...	USEFUL ANSWER
Can we start smaller?	Yes. The \$20K Starter Pod proves the model in one semester, then demo-day proof supports the \$50K or \$100K expansion.
Why not only cloud?	Cloud is useful for overflow. Local AI teaches privacy-bounded handling, predictable inference costs, and real model operations.
Who maintains this?	Use standard tools, vendor support, device management, backups, and a student peer-mentor model by month three.
How will faculty keep up?	Start with a faculty bridge and a curriculum where faculty guide projects rather than having to be expert operators on day one.

Most launch risks are manageable if they are named before procurement.

The purpose of the next memo is not only to price hardware. It should reduce operational risk for the department and make the pilot easier to defend.

RISK	MITIGATION
Procurement timing	Start with a tier that can land inside one semester. Pre-stage vendor quotes and education pricing paths.
Price volatility	Treat hardware numbers as planning figures until quote lock. Add contingency for memory, shipping, cabling, and install movement.
Arm vs x86 fragmentation	Use DGX Spark-class hardware as a shared node while keeping daily student seats broadly compatible.
Faculty adoption	Run a one-week faculty bridge, identify a second campus champion, and keep the first cohort focused on guided build modules.
Power and HVAC	Require a room electrical and cooling review for \$100K+ tiers and confirm 30-40 A circuit needs before ordering.
Weak measurement	Define pilot KPIs before the first cohort starts and report them in a semester dashboard.
Branding drift	Name and present the lab as Oakwood's from the beginning. Outside support should enable the department, not become the story.
Limited seats	Create a transparent first-cohort rubric across class year, readiness, major interest, and mentoring potential.

Risk posture. The lab does not need to solve every AI use case on day one. It needs a well-scoped first cohort, dependable tools, clear rules, and a demo day that proves the next investment is earned.

A first proof can arrive before the funding conversation cools.

The launch plan is calibrated for visible progress: diagnostic clarity, procurement readiness, baseline policy, installation, first build, and a public demo cadence.

**WEEKS 1-2****Discovery**

Confirm budget, room, network, purchasing, faculty leads, program format, and success metrics.

**WEEKS 3-4****Procure & Policy**

Pull quotes, choose the tier, draft lab use rules, create account structures, and set the first-cohort plan.

**WEEKS 5-7****Install & Baseline**

Set up workstations, shared model nodes, NAS, demo wall, backups, repositories, and first local model benchmarks.

**WEEKS 8-10****Pilot Sprint**

First cohort ships an AI-assisted app, local-model demo, evaluation report, and short portfolio recording.

**WEEKS 11-12****Demo Day**

Host the showcase, publish recap artifacts, record metrics, and identify grant, partner, and expansion opportunities.

30

days: decision memo, quotes, room plan, policy draft, cohort format.

60

days: installed baseline, accounts, storage, first local models, faculty bridge.

90

days: pilot sprint artifacts, demo day, metrics dashboard, next funding narrative.

End the meeting with a concrete deliverable and date.

The next artifact should remove uncertainty for Oakwood leadership and give the department something practical to approve, revise, or take into a funding conversation.

PROCUREMENT MEMO

Line-item vendor quotes for the selected tier, including education pricing, warranty, shipping, installation, and contingency.

SCALED ROOM LAYOUT

Room plan with seat placement, power, network, AV, storage, demo wall, and zone-by-zone occupancy.

8-WEEK PILOT CURRICULUM

Weekly module outline, deliverables, assessment rubric, faculty time estimate, and student application draft.

GOVERNANCE TEMPLATES

Lab use policy, data classification taxonomy, approval gates, prompt-injection playbook, and audit-log spec.

SUCCESS DASHBOARD

KPI definitions, baseline targets, reporting cadence, and a template for semester-one evidence.

GRANT & PARTNER NARRATIVE

A concise expansion story for NSF, NVIDIA, Microsoft, Apple education channels, Huntsville employers, and private foundations.

Recommended immediate close. If the \$50K path resonates, send a procurement memo and 8-week pilot curriculum first. If the room is uncertain, send the room layout and facilities checklist first.

Oakwood has the students, the mission, and the moment. The lab turns that into a room where students build work they can point to for the rest of their careers.